

What is NLP (Natural Language Processing)?

Natural Language Processing (NLP) is a branch of **Artificial Intelligence (AI)** that enables computers to **understand, analyze, interpret, and generate human language** such as English, Tamil, Hindi, etc.

Simple Definition

NLP helps computers communicate with humans using natural language.

Real-World Examples of NLP

1. Google Translate

Input:

Hello, How are you?

Output:

வணக்கம், நீங்கள் எப்படி இருக்கிறீர்கள்?

✅ NLP understands one language and translates it into another.

2. ChatGPT

When you ask:

What is Data Science?

ChatGPT understands your question and generates an answer.

✅ NLP is used for understanding and generating text.

3. Gmail Spam Detection

Email:

Congratulations! You won ₹1,00,000.

Output:

Spam

✅ NLP analyzes the email content and classifies it.

4. Voice Assistant

You say:

"Set an alarm for 6 AM."

The assistant understands your speech and performs the task.

✅ Speech + NLP work together.

NLP Process Flow

Human Text



Text Preprocessing



Text Analysis



Machine Learning / Deep Learning



Prediction / Result

Example

Input:

"The movie was fantastic."

Processing:

- Break sentence into words
- Understand meaning
- Analyze sentiment

Output:

Positive Review 😊

Main Applications of NLP

Application	Example
Sentiment Analysis	Positive/Negative Review
Spam Detection	Spam Email Detection
Chatbots	ChatGPT
Translation	Google Translate
Text Summarization	News Summary
Speech Recognition	Siri, Alexa
Question Answering	AI Assistants
Next Word Prediction	Mobile Keyboard Suggestions

Types of NLP Tasks

1. Text Analysis

Understanding the text.

Example:

"Chennai is the capital of Tamil Nadu."

Output:

- Chennai → City

- Tamil Nadu → State
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2. Prediction Based NLP

Predicting something from text.

Example:

"This product is amazing."

Output:

Positive Sentiment

Why NLP is Important?

Without NLP:

- Computers understand only numbers and code.

With NLP:

- Computers understand human language.
 - Humans can communicate naturally with machines.
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Interview Answer (2 Marks)

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, analyze, and generate human language. It is used in applications such as chatbots, translation systems, sentiment analysis, spam detection, and voice assistants.

One-Line Answer

NLP is the technology that helps computers understand and communicate in human language.

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NLP (Natural Language Processing) has two major types of processes:

1. Prediction Based NLP

👉 Here, the computer **predicts or classifies something** from the text.

Real-world Example 1: Spam Detection

Input Message:

"Congratulations! You won ₹10,000. Click here."

Prediction:

Spam 

Real-world Example 2: Sentiment Analysis

Customer Review:

"The food was excellent and service was fast."

Prediction:

Positive Review 

Real-world Example 3: Next Word Prediction

When you type:

"Good Morning"

Your phone suggests:

"Everyone"

The model predicts the next word.

Output

Prediction-based NLP gives a result such as:

Input	Output
Email	Spam / Not Spam
Review	Positive / Negative
News	Sports / Politics
Chat Message	Next Word Prediction

2. Text Analysis NLP

👉 Here, the computer **understands and analyzes the text** instead of predicting a category.

Real-world Example 1: Extracting Names

Sentence:

"Ratan Tata founded Tata Group."

Analysis Result:

Entity	Type
Ratan Tata	Person
Tata Group	Organization

Real-world Example 2: Word Count

Text:

"Data Science is the future."

Analysis:

- Total Words = 5
- Most Frequent Word = Data Science

Real-world Example 3: Language Translation

Input:

"How are you?"

Analysis & Conversion:

"நீங்கள் எப்படி இருக்கிறீர்கள்?"

Output

Text Analysis NLP gives information such as:

Input	Output
Document	Keywords
Sentence	Entities (Person, Place)
Paragraph	Summary
Text	Translation

Easy Difference

Prediction Based	Text Analysis
Predicts something	Understands something
Output is a label/category	Output is information/insight
Spam Detection	Keyword Extraction

Sentiment Analysis

Text Summarization

Next Word Prediction

Translation

Simple Tamil Explanation

Prediction Based NLP

"இந்த text நல்லதா? கெட்டதா? Spam-ஆ?" என்று கணிக்கிறது.

Example:

"This movie is awesome" → Positive Review

Text Analysis NLP

"இந்த text-ல என்ன தகவல் இருக்கு?" என்று ஆராய்கிறது.

Example:

"Chennai is the capital of Tamil Nadu"

Analysis:

- Chennai → Place
- Tamil Nadu → State

Interview Answer (1 line)

"Prediction-Based NLP predicts or classifies text, whereas Text Analysis NLP extracts meaning, information, and insights from text."

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